



PUBLIC SECTOR FINANCING NEEDS AND THE MULTI-YEAR PLAN MATURITY INDEX IN THE MUNICIPALITIES OF PARANÁ

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Abstract: This article examines the relationship between public sector financing needs and municipal budgetary planning capacity in the State of Paraná, as measured by the Multi-Year Plan Maturity Index (MYP-MI). The central hypothesis is that the effectiveness of fiscal adjustment depends not only on economic and fiscal variables, but also on the quality of public planning institutions. To test this hypothesis, a panel Probit model is estimated for 399 municipalities over two MYP cycles (2018–2021 and 2022–2025), including interaction terms between planning maturity and key fiscal variables, such as expenditure flexibility and dependence on intergovernmental transfers. The results indicate that reliance on the Municipal Participation Fund (MPF) is associated with a lower probability of debt reduction, while municipalities with greater economic relevance in the state context exhibit a greater capacity for fiscal adjustment. The main finding of the study is that budget planning maturity acts as a moderating factor, enhancing the effects of expenditure flexibility and mitigating the adverse effects of transfer dependence.

Keywords: Public debt; Budgetary planning; Municipalities in Paraná

Resumo: Este artigo investiga a relação entre a necessidade de financiamento do setor público e a capacidade de planejamento orçamentário nos municípios paranaenses, mensurada pelo Índice de Maturidade do Plano Plurianual (IM-PPA). A hipótese central é que a efetividade do ajuste fiscal depende não apenas de variáveis econômicas e fiscais, mas também da qualidade das instituições de planejamento. Para testar essa hipótese, estima-se um modelo *Probit* com dados em painel para 399 municípios ao longo de dois ciclos do PPA (2018–2021 e 2022–2025), incorporando interações entre o grau de maturidade do planejamento e variáveis fiscais relevantes, como flexibilidade dos gastos públicos e dependência de transferências intergovernamentais. Os resultados indicam que a dependência do Fundo de Participação dos Municípios (FPM) está associada a menor probabilidade de redução do endividamento, enquanto municípios com maior participação econômica apresentam maior capacidade de ajuste fiscal. O principal achado do estudo é que a maturidade do planejamento orçamentário atua como um fator moderador, potencializando os efeitos da flexibilidade dos gastos e atenuando os impactos adversos da dependência de transferências.

Palavras-chave: Dívida pública. Planejamento Orçamentário. Municípios paranaenses.

1. INTRODUCTION

The fiscal and economic heterogeneity among Brazilian municipalities represents one of the main challenges to the sustainability of public finances and federative coordination. Municipalities with a larger share of national gross domestic product (GDP) tend to have greater tax collection autonomy, while those with less economic dynamism remain highly dependent on intergovernmental transfers, reflecting the structural inequalities of Brazilian fiscal federalism. In this context, the combination of high budgetary rigidity and dependence on external revenues imposes significant constraints on the fiscal adjustment capacity of subnational entities.

International literature has advanced in understanding these dynamics by highlighting that fiscal sustainability does not depend solely on the level of revenue or expenditure, but also on the quality of fiscal institutions and the design of intergovernmental arrangements. The second-generation theory of fiscal federalism emphasizes that intergovernmental transfers can generate ambiguous incentives: while they may promote regional equalization, they may also induce opportunistic behavior and reduce local fiscal effort (Oates, 2005; Weingast, 2009). Recent evidence reinforces this ambiguity, indicating that reliance on transfers can compromise fiscal discipline and allocative efficiency, especially in contexts of low institutional capacity (OECD, 2021).

In addition, contemporary literature has highlighted the role of expenditure composition and tax structure in the fiscal dynamics of subnational governments. Current expenditure, particularly those associated with payroll and the provision of public services, has high institutional rigidity, limiting the ability to promote short-term adjustments (Dafflon and Vaillancourt, 2022). On the other hand, revenue composition also plays a relevant role, since more volatile tax bases, such as those associated with economic activity, tend to increase fiscal vulnerability, while property taxes contribute to greater stability and predictability of revenues (Norregaard, 2013; OECD, 2021). In this sense, fiscal sustainability results from the interaction between budget structure, economic cycle, and institutional quality.

In the Brazilian case, the 1988 Constitution promoted a significant decentralization of resources but maintained a high tax concentration in the Union, resulting in a persistent vertical fiscal gap (Politi and Mattos, 2013). Municipalities, although benefiting from constitutional transfers such as the Municipal Participation Fund (MPF), face increasing spending pressures and limitations to budget flexibility, which reduces their capacity for fiscal adjustment. Empirical evidence indicates that this dependence can discourage tax collection effort and reinforce patterns of fiscal accommodation (Hollenbach; Silva, 2019).

In this context, the role of budget planning institutions gains relevance. Studies show that the quality of public planning is a determinant factor for the intertemporal coordination of fiscal policies, contributing to greater efficiency in the allocation of resources and to the sustainability of indebtedness (OECD, 2021; Alesina; Favero; Giavazzi, 2019). However, despite advances in the international literature, empirical evidence on how planning capacity influences fiscal dynamics at the subnational level is still scarce, especially in countries with high institutional heterogeneity such as Brazil.

In the state of Paraná, the Multi-Year Plan Maturity Index (MYP-MI), developed by the State Court of Auditors, is a relevant innovation in measuring the technical and institutional quality of budget planning processes in municipalities. The indicator evaluates dimensions such as consistency of goals, alignment between budgetary instruments, transparency and social participation, offering a comprehensive measure of public planning capacity.

This article aims to investigate the impact of the maturity of budget planning on the dynamics of indebtedness in the municipalities of Paraná. It is based on the hypothesis that the effectiveness of fiscal adjustment does not depend exclusively on economic and fiscal variables,

but is conditioned by the quality of planning institutions. In particular, it is argued that performance-budget planning acts as a moderating factor, enhancing or attenuating the effects of variables such as the flexibility of public spending and the dependence on intergovernmental transfers.

To this end, the empirical strategy is based on the estimation of a panel *Probit* model, in which the probability of reducing the financing need of the municipalities is evaluated by means of the interaction between fiscal variables and public planning variables, particularly the degree of maturity of the MYP. By explicitly incorporating the institutional dimension into the analysis, the article contributes to the literature by showing that fiscal sustainability at the subnational level is the result not only of economic conditions, but also of the quality of planning and the capacity to coordinate public policies. The results suggest that the quality of public planning, measured by the MYP-MI, not only reflects institutional capacities, but also conditions the effectiveness of fiscal adjustment in the municipalities.

The article is organized into four sections, in addition to this introduction. The second section analyses the municipal budget constraint to identify relevant fiscal parameters for a municipal budget structure. The third section presents the data and the empirical strategy. The fourth section discusses the results, and the fifth concludes the paper.

2. MUNICIPAL BUDGET CONSTRAINT

The fiscal sustainability of federative entities is largely evaluated on the government's budget constraint, according to which the dynamics of public debt depend on debt service and the primary balance. This formulation, consolidated in the international literature (Blanchard, 1990; Bohn, 1998) and adapted to the Brazilian context by Mora and Giambiagi (2007) and Rocha and Rocha (2008), constitutes the starting point of this study.

Assuming that the public debt of federative entity i evolves over time, its intertemporal dynamics can be expressed as:

$$B_{it} = (1 + r_t) \cdot B_{i,t-1} - (T_{it} - G_{it}) \quad (1)$$

where B_{it} represents the debt in period t ; r_t is the interest rate on the previous accumulated debt; and T_{it} and G_{it} correspond, respectively, to primary revenue and expenditure.

Equation (1) shows that the debt trajectory is determined by the inherited accumulated debt, the cost of financing, and the primary balance. However, in the Brazilian federative context applied to the municipalities, both revenue and expenditure have relevant specificities: (i) revenues are composed of local tax efforts and intergovernmental transfers; (ii) expenditure is strongly rigid, mainly due to constitutional requirements and the high proportion of mandatory expenses.

To make this dynamic explicit, and adapted to the municipalities, it is assumed that total revenue is divided between own revenues and the revenues from transfers, while expenditure depends on the level of municipal economic activity, according to equations (2) and (3) below:

$$T_{it} = t_{it}^P \cdot Y_{it} + t_{it}^T \cdot Y_t \quad (2)$$

$$G_{it} = \theta_{it} \cdot Y_{it} \quad (3)$$

in which Y_{it} is the economic activity of municipality i ; Y_t is the economic activity of the higher entity (State or Union); t_{it}^P is the sensitivity of own revenues in relation to the municipal GDP; t_{it}^T is the sensitivity of transfers to the national economic activity; and θ_{it} is the sensitivity of expenditure to the municipal economic activity, associated with the degree of budgetary rigidity.

Substituting (2) and (3) into (1), the resulting equation is as follows:

$$B_{it} = (1 + r_t) \cdot B_{i,t-1} - [(t_{it}^P - \theta_{it}) \cdot Y_{it} + t_{it}^T \cdot Y_t] \quad (4)$$

This indicates that the debt in period t depends not only on the cost of carrying past debt, but also on the capacity to generate own revenues, the magnitude of transfers and the degree of expenditure rigidity.

To approximate the model to the metric used by the Brazilian Fiscal Responsibility Law (FRL), divide (4) by the total revenues T_{it} , obtaining a formulation in terms of the debt/revenue ratio:

$$\frac{B_{it}}{T_{it}} = \frac{(1+r_t) \cdot B_{i,t-1}}{T_{it}} - \frac{(t_{it}^P - \theta_{it}) \cdot Y_{it}}{T_{it}} - t_{it}^T \cdot \frac{Y_t}{T_{it}} \quad (5)$$

From algebraic manipulations, introducing the municipal revenue/GDP ratio, $t_{it} = \frac{T_{it}}{Y_{it}}$, the revenue growth rate, $\rho = \frac{T_{it}}{T_{i,t-1}}$, and the relative participation of the municipality in the state economy, $\varphi_{it} = \frac{Y_{it}}{Y_t}$, we arrive at:

$$\Delta b_{it} = (r_t - \rho) \cdot b_{i,t-1} - \left(\frac{1}{t_{it}}\right) \cdot \left[t_{it}^P - \theta_{it} + t_{it}^T \cdot \left(\frac{1}{\varphi_{it}}\right)\right] \quad (6)$$

where $b_{it} = \frac{B_{it}}{T_{it}}$ represents debt as a proportion of revenues and Δb_{it} is its variation.

Equation (6) makes it possible to define the public sector financing needs (PSFN) in two components:

$$\Delta b_{it \text{ PSFN}} = (r_t - \rho) \cdot b_{i,t-1 \text{ Debt coefficient}} - \left(\frac{1}{t_{it}}\right) \cdot \left[t_{it}^P - \theta_{it} + t_{it}^T \cdot \left(\frac{1}{\varphi_{it}}\right)\right] \text{ Fiscal effort = Primary balance} \quad (7)$$

According to equation (7), the variation in debt as a proportion of revenues, Δb_{it} , or the public sector financing needs (PSFN), is determined by two components: the cost of interest, or debt coefficient — the difference between the cost of debt and the growth of revenues, and the fiscal effort, represented by the government's primary balance. This effort is determined by five parameters: t_{it} , t_{it}^P , θ_{it} , t_{it}^T , and φ_{it} .

As the parameter t_{it} expresses the relationship between revenues and economic activity of the municipal entity, higher levels in this relationship reduce the impact of the fiscal effort on the PSFN. This result, consistent with the Laffer Curve, indicates the existence of an optimal tax rate that maximizes revenue without compromising the economic base. Such movement reflects the structural limits to the expansion of the primary balance of the federative entity and, consequently, to the improvement of solvency indicators, suggesting that the increase in revenues alone does not guarantee greater payment capacity.

The parameter t_{it}^P represents the component of the primary balance associated with own revenues and its impact on the PSFN. In light of the theory of fiscal federalism, the composition of these revenues — between distortive and non-distorting taxes — conditions the capacity to generate surpluses and the control of indebtedness. Municipalities that depend on narrow-based taxes face lower revenue elasticity, while those with greater tax diversification have greater fiscal stability and adjustment capacity.

The effect of the parameter θ_{it} , which measures the sensitivity of public spending to variations in local economic activity, reflects the challenges imposed by budget rigidity. Budgets with greater flexibility tend to make the fiscal effort more effective in stabilizing or reducing the PSFN, as they allow for faster adjustments in the face of economic fluctuations. Therefore, the degree of budgetary rigidity directly impacts the federative entity's ability to reallocate public spending.

The effect of transfers on the primary balance is captured by the parameter t_{it}^T . It is expected that transfers have a negative effect on municipal public debt. However, in contexts of strong fiscal dependence, this effect can be reversed, by discouraging own collection and increasing budgetary vulnerability. This dynamic especially affects indebtedness, as less predictable transfers reduce financial autonomy and hinder sustainable adjustments to public accounts.

Regarding the parameter φ_{it} , its meaning reflects the influence of the weight of municipal economic activity in the state economy. The greater the participation of the economy of the i -th municipality in the state economic activity, the greater the effectiveness of the fiscal effort. In other words, the larger the municipality's economy, the greater its tax base and, therefore, the greater its ability to promote budget adjustments via its own revenues.

Therefore, the analysis of the budget constraint of the municipalities, adapted to a federative structure, pointed out five parameters that explain non-linear effects on the relationship between debt and primary balance. Next, the validity of the hypothesis of non-linearity between debt and fiscal effort will be empirically investigated based on such five parameters.

3. EMPIRICAL STRATEGY AND DATA

This research uses equation (7) to evaluate the nonlinear relationship between debt and primary balance based on the parameters that make up the primary balance of municipal governments, t_{it} , t_{it}^P , θ_{it} , t_{it}^T , and φ_{it} . The systematization of the cause/effect relationships presented in equation (7) is the starting point for the specification of the model based on the following function:

$$\Delta b_{it} = f(t_{it}, t_{it}^P, \theta_{it}, t_{it}^T, \varphi_{it}) \quad (8)$$

In equation (8), Δb_{it} is the dependent variable, representing the variation in public debt or public sector financing needs in municipality i in year t . The variable t_{it} is the relationship between revenues and economic activity in municipality i in year t . The variable t_{it}^P represents the component of the primary balance associated with own revenues. The effect of local economic activity on local public spending is represented by θ_{it} , which reflects the challenges posed by budget rigidity and can be expressed based on an indicator of public spending flexibility, such as the ratio of capital spending to current spending. The variable t_{it}^T captures the effect of transfers on the municipal public debt. And the variable φ_{it} captures the weight of municipal economic activity in the state economy.

Additionally, since we are interested in the variation of public debt in the municipalities of Paraná, consider the following equation for a binary variable:

$$\Delta b_{it} = \begin{cases} 1 & \text{if there is a negative variation on municipal debt} \\ 0 & \text{if there is a positive variation on municipal debt} \end{cases} \quad (9)$$

The probability function for binary variables is given by

$$f(\Delta b) = p^y(1 - p)^{1-y}, \Delta b = 0, 1 \quad (10)$$

where p is the probability of y being equal to 1. This discrete random variable has the expected value $E(\Delta b) = p$

Which factors explain the probability of a decrease in municipal debt? Based on equation (8), and given the municipal budget constraint presented in Section 2, there are five quantitative explanatory variables. In addition, another qualitative variable will be devised, related to the MYP-MI, as follows:

$$MYP - MI_{it} = \begin{cases} 1 & \text{if the MYP - IM for the municipality is greater than 60\%} \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

This dummy variable of the MYP-MI assumes value 1 for municipalities that have a planning maturity index greater than 60%, therefore being classified as having high planning capacity, or value 0 otherwise. Such specifications make it possible to distinguish between municipalities regarding their public planning capacity.

Additionally, to capture non-linear effects and potential institutional complementarities, the MYP maturity variable is incorporated into the model through its interaction with two structural variables of municipal public finances. The first refers to the degree of flexibility of spending, measured by the ratio between capital expenditure and current expenditure, interpreted as an indicator of the discretionary margin for public budget allocation. The second concerns budgetary dependence, given by the ratio between transfers from the Municipal Participation Fund (MPF) and total current revenue, capturing the degree of municipal fiscal autonomy.

The inclusion of these interactive terms allows us to assess whether the effect of budget planning maturity on the fiscal variables of interest — notably the need for financing and debt dynamics — is conditioned by the level of allocative flexibility and the dependence on intergovernmental transfers. In formal terms, this specification makes it possible to identify whether municipalities with greater planning capacity (a higher MYP-MI) provide differentiated responses in contexts of greater budgetary rigidity or high fiscal dependence.

In this way, the empirical strategy adopted not only isolates the direct effect of the quality of budget planning, but also explores its conditional effects, contributing to a more comprehensive analysis of the interactions between budgetary institutions and fiscal performance at the subnational level.

To maintain the probability of choice, p , inside the range $[0,1]$, a *Probit* function specifies the standardized distribution, as demonstrated by Dinardo (2007). If Z is a standardized normal random variable, then its probability density function is

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-0,5z^2} \quad (12)$$

Whereas the *Probit function* is

$$f(z) = P[Z \leq z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-0,5u^2} du \quad (13)$$

Equation (13) represents the probability that a standardized normal random variable is to the left of the z -point. Thus, the *Probit model* expresses the probability p of y being equal to 1 from the following specification:

$$p = P[Z \leq \beta_1 + \beta'x'] = F(\beta_1 + \beta'x') \quad (14)$$

where F is the *Probit* function and x' is the vector that represents a set of explanatory variables to the public sector financing needs. The *Probit* model is nonlinear because equation (14) is a nonlinear function of β_1 and β' . Since these coefficients are unknown, we have to estimate them to find the probability and verify the effect of a one-unit change in any of the explanatory variables on the probability of $y = 1$, that is, the marginal effects. The marginal effects, considering equation (14), are represented by the following derivative:

$$\frac{dp}{dx'} = \frac{dF(t)}{dt} \cdot \frac{dt}{dx'} = f(\beta_1 + \beta'x')\beta' \quad (15)$$

where $f(\beta_1 + \beta'x')$ is the standardized normal probability density function.

Equation (15) shows the effect of an increase in one of the explanatory variables on the probability, so that the effect depends on the slope of the *Probit function*, given by $f(\beta_1 + \beta'x')$.

Chart 1 describes all the variables and their data sources. The data used in this survey were previously deflated to constant prices in 2025, based on the Extended National Consumer Price Index (ENCPI or IPCA in Portuguese), in order to ensure the intertemporal comparability of fiscal variables and eliminate the effects of inflation on the estimated results.

The analytical period was structured in line with the institutional logic of the Multi-Year Plan (MYP) and with the methodology of the MYP Maturity Index (MYP-MI), developed by the Court of Auditors of the State of Paraná (CAS-PR). Therefore, two complete budget planning cycles were considered: the first comprising the interval from years 2018 to 2021, and the second covering the period from 2022 to 2025.

The MYP-MI evaluates the quality and degree of maturity of the MYP preparation process in the municipalities of Paraná over a four-year cycle. Its methodology classifies municipal entities into five performance strata: (a) initial, (b) incipient, (c) intermediate, (d) developed, and (e) improved. Based on this classification, the main qualitative variable of the research considered as indicative of high planning capacity the municipalities with a maturity index greater than 60%, corresponding to the "developed" and "improved" strata. Otherwise, municipalities that fall below such threshold were classified as having a lower budget planning maturity.

Given that the MYP-MI reflects the quality of planning over a complete four-year cycle, it was necessary to make the other research variables compatible with this same temporal period. Thus, the fiscal and institutional variables were aggregated by means of averages calculated within each four-year cycle, in order to more adequately reflect the dynamics underlying the multi-year planning and avoid inconsistencies resulting from annual frequency analyses.

Consequently, the resulting database has two observations per municipality, corresponding to the two MYP cycles considered. Given that the State of Paraná is comprised of 399 municipalities, the total number of observations in the sample is 798. This methodological procedure allows aligning the data structure with the nature of the central indicator of the analysis, ensuring greater coherence between the measurement of the budget planning capacity and the explanatory variables associated with the fiscal performance of the municipal entities.

Table 1: Description of the variables for the *Probit model*

Variable	Description	Source
Δb_{it}	Binary variable that indicates the variation of net consolidated debt as a proportion of net current revenue. It assumes value 1 for municipalities that experience a negative change in the variable, and 0 otherwise.	National Treasury
t_{it}	Continuous variable that indicates the revenue to economic activity ratio in municipality i in year t . It considers the ratio between the municipality's current revenue and the municipal GDP.	National Treasury and Brazilian Institute of Geography and Statistics
t_{it}^P	Binary variable that indicates the composition of own revenue. Particularly, it represents the profile of the municipalities' own revenues based on the ratio between Service Tax revenue and Urban Property Tax revenue. It assumes value 1 if the municipality has a value greater than 1, and 0 otherwise.	National Treasury
θ_{it}	Continuous variable that indicates the municipal public expenditure flexibility. It is measured by the capital expenditure to current expenditure ratio.	National Treasury
t_{it}^T	Continuous variable that indicates budget dependence. It is measured by the MPF to current revenue ratio, that is, the relative participation of the Municipal Participation Fund transfers in the total current revenue of the municipality.	National Treasury
φ_{it}	Continuous variable that indicates participation of municipal GDP in the state economy. It is measured by the municipal GDP to state GDP ratio.	Brazilian Institute of Geography and Statistics
$DummyMYP60$	Binary variable that indicates the Multi-Year Plan Maturity Index. It assumes value 1 if the municipal index is greater than 60%, and 0 otherwise.	CAS-PR
$\theta_{it} DummyMYP60$	Interaction between the flexibility of public expenditure (θ_{it}) and a MYP-MI greater than 60%	National Treasury and CAS-PR
$t_{it}^T DummyMYP60$	Interaction between budget dependence (t_{it}^T) and a MYP-MI greater than 60%.	National Treasury and CAS-PR
$t_{it}^P DummyMYP60$	Interaction between the composition of own revenues (t_{it}^P) and a MYP-MI greater than 60%.	National Treasury and CAS-PR
$CurrentRevenue_{it}$	Continuous variable that indicates the natural logarithm of the current municipal revenue of municipality i in year t . Works as a control variable.	National Treasury

Source: the authors.

3.1 THE MULTI-YEAR PLAN MATURITY INDEX

The Multi-Year Plan Maturity Index (MYP-MI) was designed by the Court of Auditors of the State of Paraná (CAS-PR) to measure the degree of maturity attributable to the formalization of multi-year planning in the municipalities of Paraná. It differs from other planning metrics, such as the i-PLAN, calculated by the Rui Barbosa Institute, because it does not observe state capacities, but rather formalized planning itself.

Its conception was inspired by the performance-based budgeting model of the International Monetary Fund (IMF), which defines the essential information of a program-budget structure. The basic structure of a programmatic budget must contain: a) program objectives; b) services that the program offers; c) how the program intends to achieve its stated objectives (e.g. activities, projects, etc.); d) key performance indicators; and e) financial resources involved (Robinson and Last, 2010).

It should be noted that several national studies have already sought to assess the structuring degree of multi-year planning and its relationship with other financial variables (De Sousa et al., 2021; Lima et al., 2020; Pereira et al., 2023). However, all these studies share some methodological limitations: they consider observations about a few municipalities, do not offer longitudinal comparisons, and are subject to the subjectivity of the researcher's classification.

The MYP-MI advances in this challenge by allowing a large-scale comparison, covering all 399 municipalities of the State of Paraná in a longitudinal perspective, with a low degree of subjectivity. This is because the index is calculated for the totality of the municipalities of Paraná based on self-declared information sent by these entities through the Municipal Information System (MIS) managed by the CAS-PR, which has received structured data since 2012.

Based on the IMF's theoretical model and considering the main flaws pointed out by the literature in the process of formalizing the MYP, seven metrics were conceived, which are then analyzed together to construct the Multi-Year Plan Maturity Index (MYP-MI). Chart 2 below presents these indicators.

The relevance of the MYP-MI for the municipal fiscal dynamics can be understood in light of the budget constraint presented in section 2, regarding the determinants of fiscal effort and public sector financing needs (PSFN). As discussed, PSFN are conditioned by parameters associated with the composition of revenues, the expenditure rigidity, and the dependence on intergovernmental transfers. In this context, the MYP-MI can be interpreted as an institutional factor that affects how these three components operate and interact in the budgeting process.

Regarding budget dependence, measured by the MPF transfers to current revenue ratio, multi-year planning plays a role in the efficient allocation of resources. Municipalities with greater planning maturity tend to have a greater capacity to link transfers to strategic objectives and measurable goals, reducing problems of inefficient allocation and mitigating incentives for opportunistic behavior associated with fiscal dependence. Thus, the MYP-MI works as an institutional mechanism that conditions the impact of transfers on the fiscal effort and, consequently, on the PSFN.

As for the composition of revenue, captured by the relationship between Service Taxes and Urban Property Taxes, the quality of budget planning can influence the municipality's ability to structure its tax policy in a more predictable way and in line with medium-term fiscal objectives. Although the MYP-MI does not directly change the tax base, it can contribute to a better forecast of revenue and to the compatibility between tax collection and expenditure planning. In this sense, the planning index is associated with the ability to internalize budgetary constraints and reduce fiscal volatility resulting from tax bases that are more sensitive to the economic cycle.

Table 2: Composition of the Multi-Year Plan Maturity Index

Metric	Formulation	Description
Financial Programming (I_1)	$\frac{\text{Planned expenditure}}{\text{Revenue in previous year}}$	I_1 aims to identify whether there was a programming of expenditure in the MYP in amounts that are at least equivalent to the amount routinely executed in the previous annual budget. The upper limit of the indicator is 100%. An insufficient financial projection can

		imply difficulties in carrying out what was planned.
Core Programs (I_2)	$\frac{\text{Core programs}}{\text{Support programs}} \\ 0,7$	I_2 aims to identify whether there is proportionality between core finalistic programs and management support programs. For municipalities with a ratio equal to or greater than 70% of core programs, 100% is assigned to I_2 . For municipalities that do not have at least 70% of core programs, the proportion of core programs is calculated in relation to the fixed parameter of 70%. An excess of management support programs can indicate classification errors or lack of direction in planning.
Program Objectives (I_3)	$\frac{\text{Programs with suitable objectives}}{\text{Total of programs}}$	I_3 aims to assess the adequacy of the description provided for the objectives of the programs, penalizing the objectives with inadequate descriptions. Inadequate descriptions do not allow the public to understand the nature of the program. The assessment of suitability was processed, objectively and impartially, through an Artificial Intelligence algorithm.
Action Definition (I_4)	$\frac{\text{Programs with specific actions}}{\text{Total of programs}}$	I_4 aims to identify the proportion of programs for which actions were specified. Programs without any specified actions have very limited information, as they aren't associated either with monetary values or physical goals.
Indicator Definition (I_5)	$\frac{\text{Programs with indicators}}{\text{Total of programs}}$	I_5 aims to identify the proportion of programs for which there is at least one associated indicator. The absence of indicators does not allow for adequate monitoring of the progress of the program.
Action Measurement (I_6)	$\frac{\text{Actions with physical goals} \neq 0 \text{ or } 1}{\text{Total of actions}}$	I_6 aims to identify actions that have a realistic goal attribution. The absence of physical goals, or the presence of binary physical goals, are associated with a low quality of action measurement.
Indicator Measurement (I_7)	$\frac{\text{Indicators} \neq 0 \text{ or } 1}{\text{Total of indicators}}$	I_7 aims to identify indicators that have a realistic measurement. Indicators with a baseline measure equal 0, or binary indicators, are associated with a low quality of measurement.
MYP – MI	$\frac{I_1 + I_2 + I_3 + I_4 + I_5 + I_6 + I_7}{7}$	The MYP-MI is a composite index, formed by the arithmetic average of the 7 sub-indices that evaluate diverse elements of the quality of performance-budget planning. All sub-indices range from 0 to 1. Therefore, the general Maturity Index also varies between 0 and 100%.

Source: prepared by the authors.

Regarding the indicator of public expenditure flexibility, defined as the capital expenditure to current expenditure ratio, the MYP-MI assumes a central role in the intertemporal coordination of resource allocation. More structured planning tends to improve the quality of financial programming and the consistency between physical goals and budget execution, which can increase the ability of the public manager to adjust the intertemporal expenditure composition. Thus, the maturity of planning not only influences the level of budgetary flexibility, but also the effectiveness with which this flexibility is used to promote fiscal adjustment.

Also, to account for different qualitative stages of planning maturity, the MYP-MI, which ranges from 0 to 100%, is subdivided into 5 equidistant strata. Entities with scores in the first range (0-20%) are assigned the planning classification "Initial"; entities in the second range (21-40%), "Incipient"; entities with scores in the central range (41-60%), "Intermediate"; entities in the fourth range (61-80%), "Developed"; and finally, entities with the highest scores (81-100%), are assigned the planning classification "Improved".

Chart 3 below presents these ranges, with the number of municipalities in each category in the evaluation carried out in the 2022-2025 planning cycle.

Table 3 – Maturity strata of municipal planning

Maturity level	Range	Number of municipalities	Percentage of Municipalities
Initial	[0–20%]	14	3.51%
Incipient	(20–40%]	81	20.30%
Intermediate	(40–60%]	99	24.81%
Developed	(60–80%]	143	35.84%
Enhanced	(80–100%]	62	15.54%
Grand Total		399	100%

Source: prepared by the authors based on data from the CAS-PR.

The calculations were automated, so that it is possible to replicate the planning maturity index for past and future periods. Therefore, data were calculated for two management cycles of the Multi-Year Plan: 2018-2021 and 2022-2025. Table 4 displays the distribution of municipalities to each maturity level, as well as the difference in terms of percentage points (p.p.) in the composition of maturity levels from one cycle to the next.

Table 4: Intertemporal evolution of the maturity strata of municipal planning

Maturity level	MYP 2018-2021	MYP 2022-2025	Variation (p.p.)
Initial	2.26%	3.51%	+ 1.25
Incipient	16.54%	20.30%	+ 3.76
Intermediate	25.06%	24.81%	- 0.25
Developed	35.34%	35.84%	+ 0.50
Enhanced	20.80%	15.54%	- 5.26

Source: prepared by the authors based on data from the TCE-PR.

Overall, the MYP-MI allows for significant advancements in the research agenda that seeks to understand how the formalization of performance-budget planning contributes to generate better results in financial terms for public entities (George et al., 2019), overcoming a limitation that is commonly cited in the literature, regarding the difficulties of undertaking evaluations of this nature due to the absence of standardized, structured, and longitudinal data on planning (Poister et al., 2010).

4. RESULTS

Table 1 presents the results of the estimation of the *Probit* model, based on 798 observations, corresponding to the two MYP cycles (2018–2021 and 2022–2025) for the 399

municipalities in Paraná. The *LR Chi(2)* likelihood test indicates that the set of explanatory variables is statistically significant, allowing the rejection of the null hypothesis that all coefficients are jointly equal to zero ($Prob > Chi2=0.000$), at 1% significance level.

McFadden's pseudo-R² being equal to 0.0386 suggests a gain in the model fit in relation to the specification without any regressors, a value that is compatible with empirical applications involving discrete choice models. In addition, the model presents satisfactory predictive performance, as it correctly classified the dependent variable in 62.4% of cases.

Table 1: Estimated *Probit* results

Variables	Coefficient	Marginal effects	Standard error	Z-statistics	p-value
Constant	5,623	-	1.830	3.071	0.002***
$CurrentRevenue_{it}$	- 0.282	- 0.112	0.096	- 2.933	0.003***
θ_{it}	1.198	0.477	0.886	1.352	0.176
$\theta_{it} DummyMYP60$	- 2.575	- 1.025	1.114	- 2.312	0.020**
t_{it}^T	- 2.540	- 1.011	0.504	- 5.039	0.000***
$t_{it}^T DummyMYP60$	1.205	0.480	0.493	2.445	0.014**
t_{it}^P	- 0.249	- 0.099	0.139	- 1.793	0.073*
$t_{it}^P DummyMYP60$	0.073	0.029	0.184	0.398	0.690
t_{it}	1.410	0.561	0.936	1.506	0.132
φ_{it}	0.232	0.092	0.116	1.986	0.047**
Number of Observations	798				
McFadden's R ²	0.038640				
Percentage of correct predictions	62.4%				
LR Chi(2)	42.6681				
$Chi > Prob (2)$	0.00000				

Significant at 1%; **Significant at 5%; *Significant at 10%
Source: the authors.

It should be noted that, in the *Probit* model, the estimated coefficients do not have a direct interpretation in terms of variation in the probability of the event of interest. These coefficients represent the effect of the explanatory variables on an unobservable latent variable, associated with the municipality's propensity to present a reduction in its financing needs. In this context, the sign of the coefficients indicates only the direction of the effect—positive or negative—on this latent propensity, and not on the probability itself.

Thus, for a more economically relevant interpretation, it is necessary to analyze the marginal effects, which measure the impact of variations in the explanatory variables on the probability of occurrence of the event of interest. In the present study, such effects are calculated in relation to the probability of the municipality presenting a reduction in its net consolidated debt to net current revenue ratio.

Of the nine explanatory variables, three were not significant, while six variables were considered individually significant, according to the *Z-statistics* and the *p-value*.

The control variable, corresponding to the natural logarithm of current revenue, is statistically significant at 1%, displaying a negative marginal effect. This result indicates that

municipalities with greater revenue capacity are less likely to register a reduction in the need for financing over the period analyzed.

Although at first glance this result may seem counterintuitive, it finds support in the recent literature on subnational public finance. Evidence suggests that higher levels of revenue are often associated with more complex and rigid budget structures, characterized by high permanent spending commitments and less flexibility for short-term fiscal adjustments. Studies show that current expenditure, especially associated with payroll and the provision of public services, has high institutional rigidity, limiting the ability to respond to fiscal shocks (OECD, 2021; Dafflon and Vaillancourt, 2022).

In addition, the literature highlights that fiscal capacity does not automatically translate into better financial performance, since it is conditioned by institutional factors and the quality of budget planning. In contexts of greater availability of resources, weaknesses in planning and coordination between fiscal instruments can compromise allocative efficiency and debt sustainability (Hollenbach; Silva, 2019; OECD, 2021). In this sense, larger municipalities tend to have more complex fiscal dynamics, in which the adjustment of the budget imbalance occurs more gradually and conditioned by the expenditure structure (Bastos and Pineda, 2013).

Thus, the result suggests that a greater tax collection capacity does not guarantee, on and by itself, an improvement in fiscal indicators, and its effects are conditioned by the expenditure structure and the quality of planning institutions. Such evidence reinforces the importance of performance-budget planning as a central element for fiscal sustainability at the subnational level, thus contributing to the relevance and suitability of the MYP-MI.

The next set of variables will be analyzed based on two effects: the first, considering the degree of public spending flexibility (θ_{it}), and the second, based on the interaction between the public expenditure flexibility and the MYP-MI ($\theta_{it}DummyMYP60$). In isolation, the parameter of public expenditure flexibility was not statistically significant in the model; however, when associated with a high planning maturity, such flexibility was deemed significant at 5%. Moreover, its estimated marginal effect is negative and considerably high in absolute magnitude, indicating that, in municipalities with greater budget planning maturity, a greater capital expenditure flexibility exerts a relevant impact on the probability of reducing public sector financing needs.

Given the nonlinear nature of the *Probit* model, direct quantitative interpretation of this effect should be undertaken with caution. As highlighted by Ai and Norton (2003) and further analyzed by Greene (2018), the marginal effects in models with terms of interaction are not constant and depend on the evaluation point in the space of the covariates, so that high magnitudes do not necessarily imply variations equivalent to probability. Thus, the interpretation should privilege the sign and the statistical significance of the coefficients, as well as the economic consistency of the results.

A relevant aspect for the interpretation of this result refers to the construction of the public expenditure flexibility indicator, defined as the capital expenditure to current expenditure ratio. In principle, higher values of this indicator would be associated with greater budgetary flexibility, as they expand the discretionary margin of the public manager. However, the literature points out that the distinction between capital and current expenditure does not necessarily imply greater fiscal adjustment capacity, since both are inserted in institutional structures that condition their execution and rigidity (OECD, 2021; Dafflon; Vaillancourt, 2022).

In this context, the negative coefficient of the interaction term between spending flexibility and planning maturity suggests a more complex dynamic. Specifically, among municipalities with a higher degree of planning maturity, increases in the relative participation of capital expenditure are associated with a reduction in the probability of a decrease in the need

for financing. On the other hand, reductions in this indicator — reflecting a greater relative weight of current expenditure — are associated with a greater probability of fiscal adjustment.

A possible interpretation for this result is that municipalities with greater planning capacity tend to allocate their current expenditure more efficiently, internalizing budget constraints and prioritizing fiscal sustainability in the short term. Empirical evidence indicates that the quality of fiscal institutions and budget planning plays a central role in the intertemporal coordination of public policies and in the management of the fiscal cycle (Hollenbach and Silva, 2019; Bastos and Pineda, 2013). In addition, the literature highlights that fiscal adjustments often occur through the control of current expenditure, given their greater participation in the budget and their direct impact on the primary result (Alesina, Favero, and Giavazzi, 2019).

This finding reinforces the hypothesis that the effectiveness of budget management instruments depends on the quality of planning institutions, indicating that the expenditure flexibility, by itself, does not guarantee better fiscal results, and its effectiveness is conditioned by the capacity to coordinate and rationalize the budget process.

Regarding the effect of budget dependence, t_{it}^T , represented by the relative participation of the Municipal Participation Fund (MPF) in the total current revenue, the results indicate a negative and statistically significant coefficient at the level of 1%. This finding suggests that higher levels of dependence on intergovernmental transfers are associated with a lower probability of reducing municipal indebtedness, indicating that high fiscal dependence may limit the adjustment capacity of local public finance.

This result is consistent with the literature on fiscal federalism, which highlights that although intergovernmental transfers play a relevant role in the equalization of resources, they can generate adverse incentives for fiscal effort and budgetary discipline, especially in contexts of high fiscal dependence (Oates, 2005; Weingast, 2009; Boadway and Shah, 2007).

However, when considering the term of interaction between budget dependence and planning maturity, $t_{it}^T \text{DummyMYP60}$, a positive and statistically significant coefficient is observed at the level of 5%. This might indicate that the quality of budget planning plays a moderating role in the adverse effects of dependence on transfers, attenuating its impact on the dynamics of indebtedness. In other words, although dependence on transfers is associated with a lower probability of fiscal adjustment, this effect is partially mitigated in municipalities with greater planning maturity. This suggests that more robust planning institutions contribute to improving the allocation and use of transferred resources, reducing distortions associated with fiscal dependence.

This finding reinforces the hypothesis that the impact of intergovernmental transfers on fiscal performance is conditioned by the quality of local institutions. In line with the second-generation literature on fiscal federalism, the results indicate that the presence of more developed institutional mechanisms—such as a structured performance-budget planning—can mitigate moral hazard problems and improve fiscal discipline at the subnational level (Oates, 2005; Weingast, 2009).

The effect of the composition of own revenues, captured by the relative participation of the Service Tax in relation to the Urban Property Tax, t_{it}^P , was statistically significant at 1%. The estimated marginal effect is negative, indicating that municipalities with greater dependence on revenues from Service Taxes are less likely to reduce their financing needs. Considering the nonlinear nature of the model, this result should be interpreted in terms of the direction and significance of the parameter, suggesting that tax structures concentrated on more volatile bases —such as the services sector in the economy— may be associated with greater fiscal instability.

This finding is consistent with the literature, which highlights that taxes on services tend to be more sensitive to the economic cycle, while property taxes have a more stable and predictable basis, contributing to greater fiscal resilience of local governments (Norregaard,

2013; OECD, 2021). In this sense, the composition of tax revenue emerges as a relevant factor for municipal fiscal sustainability.

On the other hand, the interactive variable between the composition of own revenues and the maturity of the MYP, $t_{it}^P DummyMYP60$, did not show statistical significance, suggesting that, unlike what was observed for other fiscal dimensions, the quality of budget planning does not significantly alter the impact of the tax structure on the dynamics of indebtedness.

Regarding the variables associated with economic activity, the revenue to economic activity ratio, t_{it} , did not present statistical significance. On the other hand, the economic proportion of the municipality in the state GDP, φ_{it} , was statistically significant at 5%, with a positive marginal effect. This result indicates that municipalities with greater economic relevance in the state context are more likely to reduce their need for financing.

A possible interpretation for this result is that municipalities with a more dynamic economy have a greater capacity to generate their own revenues and better access to fiscal management instruments, which contributes to a greater capacity for budget adjustment. In addition, the literature suggests that more robust local economies tend to have greater tax base diversification and greater resilience to fiscal shocks, favoring the sustainability of public finances (OECD, 2021; Alesina, Favero and Giavazzi, 2019).

In general, the empirical results show that the dynamics of municipal indebtedness in Paraná is conditioned by a set of fiscal and institutional factors, whose effects are manifested in a heterogeneous and non-linear way. Particularly, this research highlights the role played by a greater maturity level of performance-budget planning, which works as a moderating element over the effectiveness of fiscal instruments. The findings indicate that, although structural variables such as the dependence on intergovernmental transfers and the rigidity in the expenditure composition are associated with a lower probability of fiscal adjustment, the presence of more robust planning institutions can mitigate such adverse effects. On the other hand, it is observed that not all fiscal channels are equally sensitive to the quality of planning, as evidenced by the lack of significance in the interaction with the composition of own revenues. Moreover, structural characteristics of the municipalities, such as their economic activity and relative insertion in the state economy GDP, are also relevant to explain the probability of reducing public sector financing needs. Taken together, the results suggest that fiscal sustainability at the subnational level does not depend solely on the availability of resources or budgetary flexibility, but also fundamentally on the institutional capacity for planning and coordinating fiscal policies.

5. FINAL CONSIDERATIONS

The objective of this article was to investigate the relationship between the public sector financing needs and the performance-budget planning capacity of the municipalities of the State of Paraná, as measured by the Multi-Year Plan Maturity Index (MYP-MI). Based on the estimation of a panel *Probit* model for two planning cycles (2018–2021 and 2022–2025), we sought to evaluate not only the direct effects of fiscal variables, but, above all, the conditioning role played by public planning institutions on the dynamics of municipal indebtedness.

The empirical results indicate that the fiscal sustainability of municipalities is influenced by structural factors, such as dependence on intergovernmental transfers, the composition of revenues and local economic insertion. In particular, the high dependence on the Municipal Participation Fund (MPF) was associated with a lower probability of fiscal adjustment, corroborating the literature that points to possible effects related to moral hazard and to the reduction of fiscal effort in contexts of strong dependence on transfers.

Nevertheless, the main finding of the study lies in the role played by the maturity of performance-budget planning. The evidence indicates that the MYP acts as a relevant

institutional mechanism in modulating the effects of fiscal variables. Specifically, it was found that the quality of planning enhances the effectiveness of the flexibility of public spending and mitigates the adverse effects of budget dependence, although it does not completely eliminate such distortions. This result suggests the existence of complementarity between fiscal instruments and institutional planning capacity, in which the effectiveness of fiscal adjustment policies depends on the quality of the planning process.

In addition, it was observed that the composition of own revenues exerts a significant influence on the local fiscal dynamics: the greater the dependence on volatile tax bases, the lower the probability of fiscal adjustment. On the other hand, the lack of significance of the interaction between this variable and the MYP-MI suggests that not all fiscal channels are equally sensitive to the quality of planning, which reinforces the selective nature of institutional effects.

From the point of view of public policy implications, the results highlight the importance of strengthening performance-budget planning instruments at the municipal level, not only as a formal requirement, but as an effective mechanism for fiscal coordination. Initiatives aimed at improving the quality of Multi-Year Plans — such as greater integration with the Budget Guidelines Law and Annual Budget Law (LDO and LOA in the original acronyms in Portuguese), clear definition of goals and indicators, and increased transparency of measurements — can contribute to the improvement of fiscal management and the sustainability of public debt. In addition, institutional capacity building policies and the strengthening of local state capacities are essential for municipalities to internalize fiscal constraints and respond more efficiently to budgetary challenges.

In conclusion, the main contribution of this article lies in the empirical demonstration that fiscal sustainability at the subnational level is not determined exclusively by economic or financial variables, but also depends on the quality of public planning institutions. Finally, some methodological and interpretative caveats should be acknowledged. First, the model relies on two MYP cycles, resulting in a relatively short time dimension. This constraint reflects the current availability of the MYP-MI data provided by the Court of Auditors of the State of Paraná. Second, there is potential endogeneity between fiscal variables and public planning variables, which is not explicitly addressed in the present specification of the model. Extending the time horizon could provide additional insights and enable the application of alternative econometric approaches to better address endogeneity concerns. Third, as discussed throughout the article, the interpretation of non-linear parameters requires caution, particularly in terms of their statistical significance and magnitude, rather than their absolute values. Notwithstanding these limitations, the research contributes to the literature by providing both theoretical and empirical evidence of the moderating role of performance-based budget planning in shaping the dynamics of public sector financing needs.

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